HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Actuarial Mathematics and Statistics

F78PB

Probability and Statistics B

IN-PERSON EXAMINATION

April Examination and Assessment Diet 2023

Duration: 2 hours

Answer all six questions

Question 1: 10 marks
Question 2: 12 marks
Question 3: 8 marks
Question 4: 9 marks
Question 5: 12 marks
Question 6: 9 marks

Total: 60 marks

Attached: Distributions formula sheet
Correlation and Regression Formula sheet

Supplied: Blue Statistical Tables
Yellow Statistical tables

Calculators are permitted

1. (a) Claim sizes in a certain insurance situation are modelled by a distribution with moment generating function $M_X(\theta)$ given by

$$M_X(\theta) = (1 - 10\theta)^{-2} \ .$$

Use the equation that links expected values to the derivatives of moment generating functions to show that $E(X^2) = 600$. Also find the value of $E(X^3)$.

(4 marks)

(b) Suppose that X is a $U(0,1)$ random variable and suppose that $Y = -\log(X)$. Let $M_Y(\theta)$ be the moment generating function of Y.

(i) Show that $M_Y(\theta) = E(X^{-\theta})$

(2 marks)

(ii) Hence, by using an appropriate integral, show that $M_Y(\theta) = \frac{1}{1-\theta}$ and so determine the distribution of Y.

(3 marks)

(iii) Determine $E((\log(X))^2)$

(1 mark)

[Total 10 marks]

2. Suppose that $\underline{X} = (X_1, X_2, ..., X_n)$ is a random sample from a distribution with density function

$$f_X(x;\theta) = \begin{cases} \frac{2x+\theta}{2\theta^2}, & 0 \le x < \theta \\ 0, & x < 0 \text{ or } x \ge \theta \end{cases}$$

(a) A statistician has proposed an estimator for θ as $\hat{\theta} = 2\bar{X}$.

(i) Calculate the bias of this estimator.

(4 marks)

(ii) Show that the variance of the distribution with the above density is $\frac{11}{144}\theta^2$ and hence calculate the Mean Squared Error (MSE) of $\hat{\theta}$.

(6 marks)

(b) By making use of some of your working from part (a), give the method of moments estimator (MME) for θ.

(2 marks)

[Total 12 marks]

3. A medical scientist is conducting a study on pregnancy duration. She has been monitoring 12 patients and is ready to complete the study and run an analysis, but 2 women are still pregnant. She decides to do the analysis now anyway using a censored data technique.

The time, T days, until a woman gives birth is proposed as being modelled by an **exponential distribution** with parameter $\boldsymbol{\beta}$. The women in the study have been monitored over the last 270 days and for those who have given birth, the sum of their pregnancy durations, $\sum t_i$ is 2673 days.

(a) Show carefully that the likelihood for these data is given by $L(\beta) = \beta^{10} e^{-3213\beta}$.

(4 marks)

(b) Find an equation for $\hat{\beta}$, the maximum likelihood estimate (MLE) of β and use it to estimate β (to 3 significant figures).

(4 marks)
[Total 8 marks]

4.

(a) 63 individuals from a random sample of 100 ten-year old children say they have played the game "Minecraft" in the last week. It is desired to obtain a confidence interval for the proportion, *p*, of children in general who have played this game in the last week.

(i) Explain why $\dfrac{\hat{P} - p}{\sqrt{\dfrac{p(1-p)}{m}}}$ is a pivotal quantity for *p.*

(Note: $\hat{P}$ is the random variable representing the estimate of population proportion and *m* is the sample size).

(3 marks)

(ii) Find an approximate 95% equal-tailed confidence interval for the population proportion of children who have played "Minecraft" in the past week.

(3 marks)

(b) The heights of ten-year old children in two random samples are measured. Sample 1 contains the results for children from Mexico and sample 2 contains the results for children from Australia. Details of the sample variances (cm^2) and sample sizes are shown below.

s_1^2	n_1	s_2^2	n_2
10	25	12	13

Find a 95% confidence interval for the ratio of population variances σ_1^2 / σ_2^2. Assume that the data are normally distributed.

(3 marks)

[Total 9 marks]

5. The anxiety levels of people with aerophobia (fear of flying) after receiving two different relaxation techniques are shown below.

Group 1	45	62	64	55	70	58	62	59	51	67
	72	61	61	50	66	78	58	66	62	54
Group 2	57	69	65	62	79	72	69	62	56	79
	65	71	56	80	67	67	84	64	69	65

The means and variances of these samples are summarised in the table below.

$\overline{x_1}$	s_1^2	$\overline{x_2}$	s_2^2
61.05	61.73	67.9	62.94

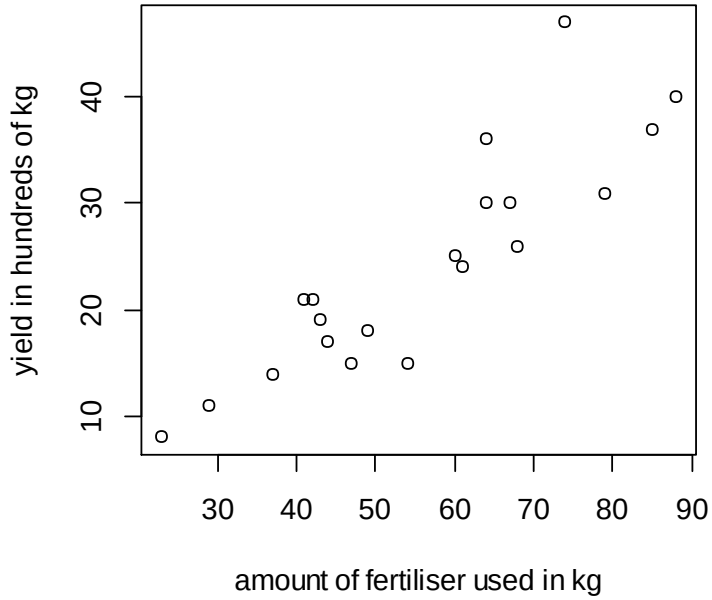
(a) By calculating a suitable test statistic and determining the relevant critical region, test the hypothesis that there is no difference between the two different relaxation techniques (at α = 0.01). Express your hypotheses and conclusion clearly. (4 marks)

(b) Calculate an approximate p-value for the test in (a), give its interpretation and state your conclusion. (3 marks)

(c) State the assumptions required for the use of this test. (2 marks)

[Total 9 marks]

6. An agricultural research organisation is testing a particular chemical fertiliser on maize to try to find out whether an increase in the amount of fertiliser used will lead to a corresponding increase in maize yield. The crop was planted in twenty equally sized plots each of which was treated with a different dose of the fertiliser (*x*) and the yield of the crop (*y*) in each plot was recorded at the end of the season. Each plot covered an area of one acre. The data for the 20 plots are shown in the following chart.



You are given the following calculations for the data shown in the graph.

$$\sum x = 1119, \sum x^2 = 68807, \sum y = 485, \sum y^2 = 13799, \sum xy = 30284$$

(a) Comment on the relationship between yield of maize (*y*) and amount of fertiliser used (*x*).

(1 mark)

(b) Determine the correlation coefficient between *x* and *y*.

(4 marks)

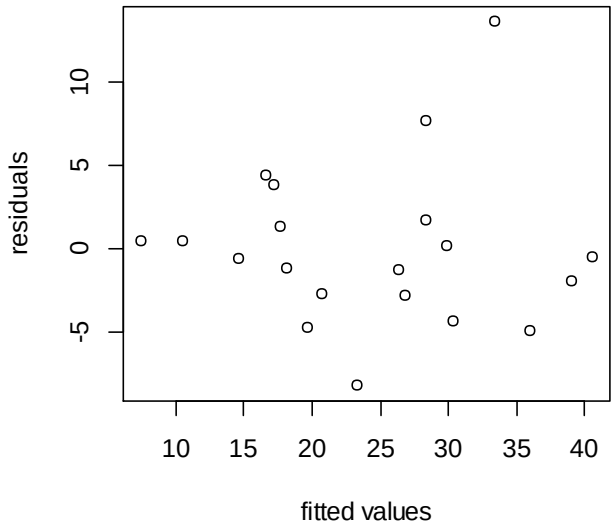
(c) Perform a statistical test to determine if the correlation coefficient is significantly different from 0.

(3 marks)

(d) Determine the parameters of a linear regression of *y* on *x* and state the fitted model equation.

(2 marks)

(e) The graph below shows the relationship between residuals and fitted values for this regression equation.



What does this graph tell you about the adequacy of the model? (2 marks)

[Total 12 marks]

END OF PAPER

Some Absolutely Continuous Distributions

Name	Notation	p.d.f.	range	m.g.f.	mean	variance
Uniform	$U(a,b)$	$\frac{1}{b-a}$	$a \le x \le b$	$\frac{e^{b\theta}-e^{a\theta}}{(b-a)\theta}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Normal	$N(\mu,\sigma^2)$	$\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$	$-\infty < x < \infty$	$e^{\mu\theta+\frac{1}{2}\sigma^2\theta^2}$	μ	σ^2
Gamma	$\Gamma(t,\lambda)$	$\frac{\lambda^t}{\Gamma(t)}x^{t-1}e^{-\lambda x}$	$0<x<\infty$, $t,\lambda>0$	$\left(\frac{\lambda}{\lambda-\theta}\right)^t$	$\frac{t}{\lambda}$	$\frac{t}{\lambda^2}$
Exponential	$\text{Exp}(\lambda)$	$\lambda e^{-\lambda x}$	$0<x<\infty$, $\lambda>0$	$\frac{\lambda}{\lambda-\theta}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Chi-square	χ^2_n	$\frac{1}{2^{n/2}\Gamma(n/2)}x^{n/2-1}e^{-x/2}$	$0<x<\infty$, $n>0$ integer	$(1-2\theta)^{-n/2}$	n	$2n$
Beta	$\beta(\alpha_1,\alpha_2)$	$\frac{\Gamma(\alpha_1+\alpha_2)}{\Gamma(\alpha_1)\Gamma(\alpha_2)}x^{\alpha_1-1}(1-x)^{\alpha_2-1}$	$0<x<1$, $\alpha_1,\alpha_2>0$	not useful	$\frac{\alpha_1}{\alpha_1+\alpha_2}$	$\frac{\alpha_1\alpha_2}{(\alpha_1+\alpha_2)^2(\alpha_1+\alpha_2+1)}$
Cauchy	–	$\frac{\lambda}{\pi(\lambda^2+(x-\mu)^2)}$	$-\infty<x<\infty$, $\lambda>0$	doesn't exist but char. fn. is $e^{i\mu\theta-\lambda\lvert\theta\rvert}$	no moments exist	
Multivariate Normal	$N(\mu,\Sigma)$	$\frac{\exp\left\{-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right\}}{(2\pi)^{d/2}\det(\Sigma)^{1/2}}$	$x\in\mathbb{R}^d$	$e^{\mu^T\theta+\frac{1}{2}\theta^T\Sigma\theta}$	μ	covariance matrix Σ

Some Discrete Distributions
(All distributions below have integer-valued ranges)

Name	Notation	prob. fn.	range	m.g.f.	mean	variance
Binomial	$\text{Bin}(n,p)$	$\binom{n}{x}p^x(1-p)^{n-x}$	$0\le x\le n$ $0<p<1$	$[(1-p)+pe^\theta]^n$	np	$np(1-p)$
Poisson	$\text{Pois}(\lambda)$	$\frac{e^{-\lambda}\lambda^x}{x!}$	$0\le x<\infty$ $\lambda>0$ real	$e^{\lambda(e^\theta-1)}$	λ	λ
Hyper-geometric	$\text{Hgeo}(n,a,b)$	$\frac{\binom{a}{x}\binom{b}{n-x}}{\binom{a+b}{n}}$	$0\le x\le n$ $a,b>0$ integers	not useful	$\frac{na}{a+b}$	$\frac{nab(a+b-n)}{(a+b)^2(a+b-1)}$
Negative binomial	$\text{Nbin}(k,p)$	$\binom{x-1}{k-1}p^k(1-p)^{x-k}$	$k\le x<\infty$, $0<p<1$	$p^k[e^{-\theta}-(1-p)]^{-k}$	k/p	$k(1-p)/p^2$
Geometric	$\text{Geo}(p)$	$p(1-p)^{x-1}$	$1\le x<\infty$ $0<p<1$	$p[e^{-\theta}-(1-p)]^{-1}$	$1/p$	$(1-p)/p^2$

Multinomial: multivariate extension of binomial

Prob. fn. is given by

$$f(x_1,x_2,\ldots,x_k)=\frac{n!}{x_!!x_2!\ldots x_k!x_{k+1}!}p_1^{x_1}p_2^{x_2}\ldots p_k^{x_k}p_{k+1}^{x_{k+1}},\quad 0\le x_i\le n,\ \ 0<p_i<1\ \ \forall i=1,2,\ldots,n+1$$

where $p_1+p_2+\cdots+p_{k+1}=1$ and $x_1+x_2+\cdots+x_{k+1}=n$.

The m.g.f. is given by

$$[p_1e^{\theta_1}+p_2e^{\theta_2}+\cdots+p_ke^{\theta_k}+p_{k+1}]^n.$$

Moments: $E(X_i)=np_i$, $\text{Var}(X_i)=np_i(1-p_i)$, $\text{cov}(X_i,X_j)=-np_ip_j$, $i,j=1,2,\ldots,n+1$.

Correlation and Regression

Basic sums:

$$S_{xy} = \sum x_i y_i - \frac{1}{n}(\sum x_i)(\sum y_i)$$

$$S_{xx} = \sum x_i^2 - \frac{1}{n}(\sum x_i)^2 \quad \text{and} \quad S_{yy} = \sum y_i^2 - \frac{1}{n}(\sum y_i)^2$$

Correlation coefficient:

$$r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}} \quad \text{and} \quad t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \sim t_{n-2} \text{ under } H_0 : \rho = 0$$

Under the assumption that (x,y) has the bivariate normal distribution, we can test $H_0 : \rho = \rho_0$ against $H_1 : \rho \neq \rho_0$ using the approximation:

$$W = \frac{1}{2}\log\frac{1+r}{1-r} \sim N\left(\frac{1}{2}\log\frac{1+\rho}{1-\rho}, \frac{1}{n-3}\right)$$

Regression parameter estimates:

$$\hat{\beta} = \frac{S_{xy}}{S_{xx}} \quad \text{and} \quad \hat{\alpha} = \frac{1}{n}\{\sum y_i - \hat{\beta}\sum x_i\}$$

$$\hat{\sigma}^2 = \frac{1}{n-2}\{S_{yy} - \frac{S_{xy}^2}{S_{xx}}\}$$

Residuals and fitted values:

Fitted values $\hat{y}_i = \hat{\alpha} + \hat{\beta} x_i$ and Residuals $\hat{e}_i = y_i - \hat{y}_i$

Slope coefficient:

$$t = \frac{\hat{\beta} - \beta}{se(\hat{\beta})} \sim t_{n-2} \quad \text{where} \quad se(\hat{\beta}) = \sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}$$

Prediction standard errors:

For <u>expected</u> Y at $X = x_0$: $\hat{\sigma}\sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$ and for <u>individual</u> Y: $\hat{\sigma}\sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} + 1}$